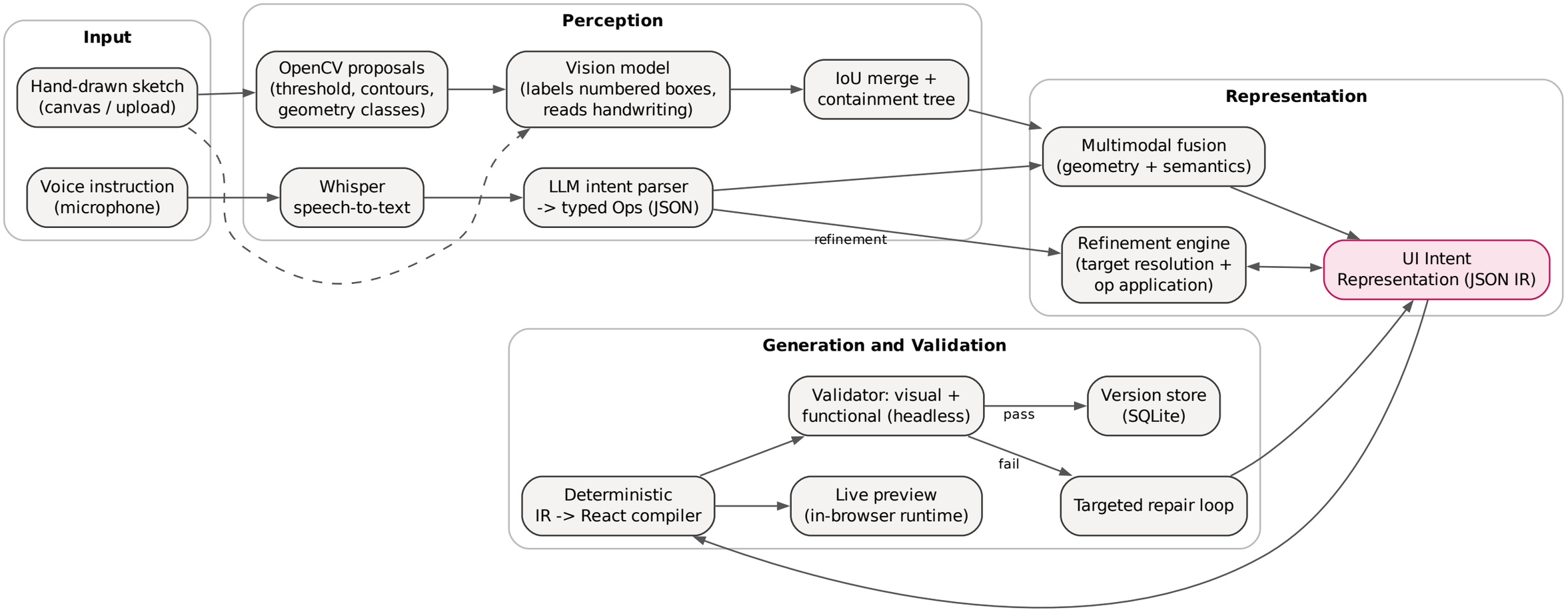
Input
Hand-drawn sketch (canvas / upload)
Voice instruction (microphone)
Perception
OpenCV proposals (threshold, contours, geometry classes)
Vision model (labels numbered boxes, reads handwriting)
IoU merge + containment tree
Whisper speech-to-text
LLM intent parser -> typed Ops (JSON)
refinement
Representation
Multimodal fusion (geometry + semantics)
Refinement engine (target resolution + op application)
UI Intent Representation (JSON IR)
Generation and Validation
Validator: visual + functional (headless)
pass
Version store (SQLite)
fail
Deterministic IR -> React compiler
Live preview (in-browser runtime)
Targeted repair loop